AFN-1164 MMA131

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

First Semester

Mathematics

ALGEBRA—I

(Non CBCS—2004 onwards)

Time : 3 Hours Maximum : 60 Marks

Section - A (6 × 5 = 30)

Answer any **six** questions.

1. If $0(G) = p^2$ where p is a prime, show that G is abelian.

2. Discuss the number and nature of the 3-sylow subgroups and 5-Sylow subgroups of a group of order $3^2.5^2$.

3. State and prove second part of Sylow's Theorem.

4. Let R be a Euclidean Ring and let A be an idea of R. Show that there exist an element $a_0 \in A$ such that A consists exactly of all $a_0 x$ as x ranges over R.

5. Let R be a Euclidean Ring. Then show that any two elements a and b in R have a greatest common divisor d and $d = \lambda a + \mu b$ for some $\lambda, \mu \in R$.

6. Prove that an ideal A = (a_0) is a maximal ideal of the Euclidean Ring R if and only if a_0 is a prime element of R.

7. If p is a prime number of the form $4n + 1$, show that the congruence $x^2 \equiv -1 \bmod p$ has a solution. Hence prove Fermat's Theorem.

8. If R is a unique factorization domain, prove that the product of two primitive polynomials in $R[x]$ is again a primitive polynomial in $R[x]$. Hence deduce that if $f(x)$ and $g(x)$ are in $R[x]$, then $c(fg) = c(f)\,c(g)$.

9. If R is a unique factorization domain and if $p(x)$ is a primitive polynomial in $R[x]$, show that it can be factored in a unique way as the product of irreducible elements in $R[x]$.

10. Show that a group G is solvable if and only if $G^{(k)} = (e)$ for some integer k.

Section - B $(3 \times 10 = 30)$

Answer any **three** questions.

11. If p is a prime number and $p/0(G)$, show that G has an element of order p.

12. For a given prime p, show that the number of p-Sylow subgroups in G is of the form $1 + kp$.

13. Show that every finite abelian group is the direct product of cyclic groups.

14. State and prove Jordan-Holder Theorem.

15. If A and B are submodules of a module M prove the following.

(a) $A \cap B$ is a submodule of M.

(b) A + B is a submodule of M.

(c) (A + B)/B is isomorphic to $A/(A \cap B)$.

16. State and prove Schreier's Refinement Theorem.

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AFN-1173 MMA332

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Third Semester

Mathematics

TOPOLOGY—I

(Non CBCS—2004 onwards)

Time : 3 Hours Maximum : 60 Marks

Section - A (6× 5 = 30)

Answer any **six** questions.

All questions carry equal marks.

1. If $\mathscr{B}$ is a basis for the topology of X and $\mathscr{C}$ is a basis for the topology of Y. Prove that the collection $D = \{ B \times C \mid B \in \mathscr{B} \text{ and } C \in \mathscr{C} \}$ is a basis for the product topology of X×Y.

2. Let Y be a subspace of X. Prove that a set A is closed in Y if and only if it equals the intersection of a closed set of X and Y.

3. State and prove intermediate value theorem.

4. Prove that a space X is locally connected if and only if for every open set $\cup$ of X, each component of $\cup$ is open in X.

5. Prove that every closed subspace of a compact space is compact.

6. Prove that every compact subspace of a Hausdorff space is closed.

7. Prove that a product of Hausdorff space is Hausdorff.

8. Prove that every compact Hausdorff space is normal.

9. Let X be completely regular. Show that X is connected if and only if $\beta(X)$ is connected.

10. Let X be a completely regular space. If Y_1 and Y_2 are two compactifications of X show that Y_1 and Y_2 are equivalent.

Section - B

$(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

11. (a) If $\mathcal{B}$ is a basis for the topology of X then prove that the collections $\mathcal{B}y = \{B \cap Y \mid B \in \mathcal{B}\}$ is a basis for the subspace topology on Y.

(b) Show that every order topology is Hausdorff.

12. Prove that the real line R is connected.

13. Prove that every closed interval in R is compact.

14. State and prove Urysohn Lemma.

15. Prove that an arbitrary product of compact space is compact in the product topology.

16. State and prove Tietze extension theorem.

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AFN-1175 MMA3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Third Semester

Mathematics

Elective : OPERATION RESEARCH

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 60 Marks

Part - A (6 × 5 = 30)

Answer any **six** questions.

1. List out the steps of the network simplex method.

2. A project consists of a series of tasks labelled A, B, ... H, I with the following relationships [W < X,Y means X and Y cannot start until W is completed ; X, Y < W means W cannot start until both X and Y are completed] With this notation construct the network diagram having the following constraints :

   A < D, E ; B, D < F; C < G ; B, G < H ; F, G < I. Draw the network for the project

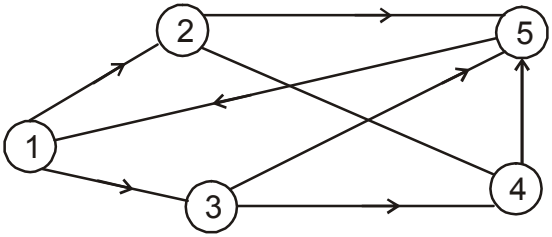
3. What are the advantages of Network Techniques ?

4. For the network given below, determine :

(a) A path. (b) A cycle.

(c) A directed cyle. (d) A tree.

(d) A spanning tree.



5. Derive the general formula for the

$(M/M/1):(GD/\infty/\infty)$ queue Model.

6. Babies are born in a sparsely populated state at the rate of one birth every 12 minutes. The time between births follows and exponential distribution. Find the following :

(a) The average number of births per year.

(b) The probability that no births will occur in one day.

(c) The probability of issuing 50 birth certificates in 3 hours given that 40 certificates were issued during the first 2 hours of the 3 hour period.

7. Prove that a necessary condition for X_0 to be an extreme point of $f(X)$ is that $\nabla f(X_0) = 0$.

8. Derive Kuhn - Tucker necessary conditions for identifying stationary points of a non - linear constrained problem.

9. Explain Direct Search Method.

10. Solve the following problem by geometric programming.

Minimize : $Z = 5x_1 x_2^{-1} + 2x_1^{-1} x_2 + 5x_1 + x_2^{-1}$

$$x_1, x_2 > 0.$$

Part B (3 × 10 = 30)

Answer any **three** questions.

11. The following table gives the activities in a construction project and time duration.

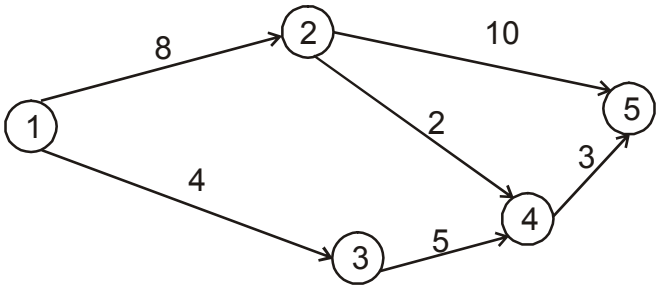
Activity	Preceding activity	Normal time (days)
1 – 2	–	20
1 –3	–	25
2 – 3	1 – 2	10
2 – 4	1 – 2	12
3 – 4	1 – 3, 2 – 3	5
4 – 5	2 – 4, 3 – 4	10

(a) Draw the activity network of the project.

(b) Find the Total float and Free-float for each activity.

(c) Determine the Critical path and the Project duration.

12. Consider the network in the following figure.



Compute the different minimum - cost schedules that can occur between normal and crash time.

Activity	Normal		Crash	
(i,j)	Duration	Cost	Duration	Cost
(1, 2)	8	100	6	200
(1,3)	4	150	2	350

Activity	Normal		Crash	
(i,j)	Duration	Cost	Duration	Cost
(2, 4)	2	50	1	90
(2, 5)	10	100	5	400
(3, 4)	5	100	1	200
(4, 5)	3	80	1	100

13. Customers arrive in a Poisson stream at a state revenue office. The rate of arrival between 8.00 A.M. and 10.00 P.M. is 20 per hour. Compute the following :

(a) The average number of arriving customers during the 2 hours period.

(b) The probability that no customers will arrive between 8.00 and 8.10.

(c) The probability that one customer will arrive between 8.10 and 8.20 given that no customers have arrive betwen 8.00 and 8.10

14. Solve by the Jacobian method

Maximize = $f(X) = 5x_1 + 3x_2$

subject to $g_1(X) = x_1 + 2x_2 + x_3 - 6 = 0$

$g_2(X) = 3x_1 + x_2 + x_4 - 9 = 0$

$x_1, x_2, x_3, x_4 \geq 0$

15. Solve by quadratic programming technique

Maximize $Z = 4x_1 + 6x_2 - 2x_1^2 - 2x_1 x_2 - 2x_2^2$

subject to $x_1 + 2x_2 \leq 2$

$x_1, x_2 \geq 0.$

16. Explain (M / M / 1) : (GD / N / ∞) model. Obtain L_s, L_q, W_s and W_q.

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AFN-1177 MMA431

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fourth Semester

Mathematics

FUNCTIONAL ANALYSIS

(Non CBCS—2004 onwards)

Time : 3 Hours Maximum : 60 Marks

Section - A (6 × 5 = 30)

Answer any **six** questions.

1. State and prove Hahn-Banach Separation theorem.

2. Let X be a normed space and Y be a closed subspace of X. Then prove that X is a Banach space if and only if Y and X/Y are Banach spaces in the induced norm and the quotient norm respectively.

3. State the prove uniform boundedness principle.

4. State and prove open mapping theorem.

5. Let x and y be normed spaces and $F \in BL(X-\ )$

Then :

(a) $Z(F) = \{x \in X : x'(x) = 0 \text{ for all } x' \in R(F')\}$

(b) $Z(F') = \{y' \in Y' : y'(y) = 0 \text{ for all } y \in R(F)\}$

In particular, F′is one-to-one iff R(F) is dense in Y.

6. Let X be a separable normed space. Then every bounded sequence in X' has a weak * convergent subsequence.

7. State and prove Schwarz inequality.

8. Let X be an inner product space. Let $\{x_1, x_2, \ldots x_n,\}$ be an orthogonal set in X. Then prove that

$$\|x_1 + x_2 + \ldots + x_n\|^2 = \|x_n\|^2 + \ldots + \|x_n\|^2.$$

9. Let H be a Hilbert space. For $f \in H'$, let y_f be the representer of f in H. Then prove that the map $T : H' \to H$ given by $T(f) = y_f$ is a surjective conjugate linear isometry.

10. Let X be an inner product space and $f \in X'$. Let $\{u_1, u_2, \ldots\}$ be an orthonormal set in X. Then prove that $\sum_n |f(u_n)|^2 \leq \|f\|^2$.

Section - B

(3 × 10 = 30)

Answer any **three** questions.

11. (a) State and prove Resonance theorem.

(b) Let X and Y be normed spaces and $F : X \to Y$ be linear. Then prove that F is continuous iff $g o F$ is continuous for every $g \in Y'$.

12. State and prove closed graph theorem.

13. Let X and Y be Banach spaces and $F \in BL(X, Y)$. Then prove that $R(F) = Y$ iff F' is bounded below.

14. State and prove Riesz representation theorem for L^P.

15. State the prove Gram-Schmidt orthonormalization theorem.

16. State and prove Hahn Banach extension theorem.

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AFN-1180 MMA4E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fourth Semester

Mathematics

Elective—MATHEMATICAL STATISTICS—II

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 60 Marks

Part - A (6 × 5 = 30)

Answer any **six** questions.

1. Let a random sample of size 17 from the normal distribution $N(\mu, \sigma^2)$ yield $\bar{x} = 4.7$, and $s^2 = 5.76$. Determine a 90 percent confidence interval for μ.

2. Let Y be b (300, p). If the observed value of Y is y = 75, find an approximate 90% confidence interval for P.

3. Explain different types of errors.

4. Let X have a p d f of the form $f(x\,;\theta) = \theta\, x^{\theta-1}$, $0 < x < 1$, zero elsewhere, where $\theta \in \{\theta : = 1, 2\}$. To test the simple hypothesis. $H_o : \theta = 1$ against the alternative simple hypothesis $H_1 : \theta = 2$, use a random sample X_1, X_2 of size $n = 2$ and define the critical region to be $C = \left\{(x_1, x_2) : \frac{3}{4} \le x_1\, x_2\right\}$. Find the power function of the test.

5. Explain the randomized test with an example.

6. In 200 tosses of a coin, 115 Heads and 85 Tails were observed. Test the hypothesis that the coin is fair using a level of significance of 0.05.

7. Let $X_1, X_2, \ldots, X_n$ represent a random sample from a distribution having the p d f

$f(x;\theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1, 0 < \theta < \infty \\ 0, & \text{elsewhere} \end{cases}$. Find the mle $\hat{\theta}$ of θ

8. Let $X_1, X_2, \ldots, X_n$ be *i i d* N $(0, \theta)$, $0 < \theta < \infty$. Prove that $\sum_{1}^{n} X_i^2$ is a sufficient statistics for θ.

9. Compute the mean and the variance of a random variable that is $\chi^2(r, \theta)$.

10. Explain Chi-square test for independence of attributes.

Part - B (3 × 10 = 30)

Answer any **three** questions.

11. State and prove Cramer-Rao inequality.

12. State and prove the factorization theorem of Neyman.

13. Let $X_1, X_2 \ldots, X_{25}$ denote a random sample of size 25 from a normal distribution N (θ, 100). Find a uniformly most powerful critical region of size $\alpha = 0.10$ for testing $H_0 : \theta = 75$ against $H_1 : \theta > 75$.

14. Let X be N (0, θ) and in the usual notation, let $\theta^1 = 4$, $\theta^{11} = 9$, $\alpha_a = 0.05$, and $\beta_a = 0.10$. Show that the sequential probability ratio test can be based upon the statistic $\sum_{1}^{n} X_i^2$. Determine $C_o(n)$ and $C_1(n)$.

15. Give the following observations in a two-way classification with $a = 3$, $b = 4$, and $c = 2$, compute the F-statistics used to test that all interactions are equal to zero ($\gamma_{ij} = 0$), all column means are equal ($\beta_j = 0$), and all row means are equal ($\alpha_i = 0$) respectively.

Row/column	1	2	3	4
1	3.1	4.2	2.7	4.9
	2.9	4.9	3.2	4.5
2	2.7	2.9	1.8	3.0
	2.9	2.3	2.4	3.7
3	4.0	4.6	3.0	3.9
	4.4	5.0	2.5	4.2

16. Describe the analysis of variance problem.

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AFN-1181 MMA4E5

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Fourth Semester

Mathematics

TOPOLOGY—II (Elective)

(Non CBCS—2004 onwards)

Duration : 3 Hours Maximum : 60 Marks

Section - A (6 × 5 = 30)

Answer any **six** Questions.

1. Let X be locally compact Hausdorff. Let Y be a subspace of X *gf* Y is closed in X or open in X then prove that Y is locally compact.

2. Let $A \subset X$; let $f : A \to Z$ be a continuous map of A into the Hausdorff space Z. Prove that there is almost one extension of *f* to a continuous function $g : \overline{A} \to Z$.

3. Let @ be a locally finite collection of subsets of X. Prove that following :–

(a) Any subcollection of @ is locally finite.

(b) The collection $B = \{\overline{A}\}_{A \in @}$ of the closures of the elements of @ is locally finite.

(c) $\overline{\bigcup_{A \in @} A} = \bigcup_{A \in @} \overline{A}$.

4. Show that the irrationals are a G_δ set in R.

5. Prove that the Euclidean space R^k is complete in either of its usual metrics, the euclidean metric d or the square metric ρ.

6. Let X be a topological space and let (Y,d) be a metric space. Prove that the set $\mathscr{C}(X,Y)$ of continuous functions is closed in YX under the uniform metric. Prove also if Y is completed then $\mathscr{C}(X,Y)$ is complete in the uniform metric.

7. Prove that a metric space $\mathscr{C}(X,d)$ is compact iff it is complete and totally bounded.

8. If X is locally compact or if X satisfies the first countability axiom then prove that X is compactly generated.

9. Show that every open subset of a Baine space is a Baine space.

10. If X is a compact Hausdorff space or a complete metric space, prove that X is a Baine space.

Section - B

$(3 \times 10 = 30)$

Answer any **three** Questions.

11. Let X be a locally compact Hausdorff space which is not compact. Let Y be the one-point compactification of X. Prove the following :–

(a) Y is a compact Hausdorff space.

(b) X is a subspace of Y.

(c) The set Y-X consists of a single point and $\overline{X} = Y$.

12. Let X be a regular space with a basis B that is locally countable locally finite. Prove that X is metrinzable.

13. Let X be a metrizable space. gl @ in an open covering of X, prove that there is a collection $\mathscr{D}$ of subsets of X s.t .

(a) $\mathscr{D}$ is an open covering of X.

(b) $\mathscr{D}$ is a refinement of @.

(c) $\mathscr{D}$ is a countably locally finite.

14. State and prove Ascoli's theorem.

15. Let X be a space. Let (Y,d) be a metrix space. For the function space Y^x, prove that the following inclusion of topologies hold.

(Uniform) $\supset$ (Compact convergence) $\supset$ Pointwise convergence)

16. Let X be a locally compact Hausdorff space. let $\mathscr{C}(x,y)$ have the compact open topology. Let Z be an arbitrary topological space. Prove that the map $F: XxZ \to Y$ is continuous iff the induced map $\hat{F}: Z \to \mathscr{C}(X,Y)$ is continuous, where $\hat{F}$ is defined by the rule $\hat{F}(z)(x) = F(x,z)$.

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